The Gregorian Calendar, Leap Years and Clock Time (IST)

What you will be able to do

1. Explain why a calendar is hard to design: days, months and years do not divide evenly.
2. Explain why there are **12 months** and where the **30- and 31-day** month lengths came from.
3. Apply the **Gregorian leap-year rule** and explain why it is needed.
4. Explain why 10 days "disappeared" in 1582 and 11 days in 1752.
5. Calculate **local time versus Indian Standard Time** from longitude.

1. The calendar-maker's headache

Natural cycle	Length
Day (Sun to Sun)	1 day
Month (Moon's phases)	29.530,59 days
Year (seasons — tropical year)	365.242,19 days

None divides evenly into the others: a year is 12.37 lunar months, or 365.24 days. Every calendar is a compromise:

- **lunar** calendars (Islamic, Lecture 1.5) follow the Moon and ignore the seasons;
- **lunisolar** calendars (Hindu, Lecture 1.4; also Jewish and Chinese) follow the Moon but add leap months;
- **solar** calendars (Gregorian, Indian National, Persian) follow the seasons and ignore the Moon.

Why 12 months? Because a year holds about 12 lunations ($365.24/29.53 = 12.37$). The word *month* comes from *moon* (compare Sanskrit *māṣ*). The Gregorian calendar keeps the number 12 but no longer follows the real Moon.

2. From the Roman calendar to Julius Caesar

The 10-month calendar of Romulus (about 753 BCE)

The earliest Roman calendar had only **10 months (304 days)**, starting in **March**; the ~61 winter days were simply not counted!

#	Roman month	Meaning	Today
1	Martius	Mars (god of war)	March
2	Aprilis	(to open?)	April
3	Maius	Maia (a goddess)	May
4	Iunius	Juno	June

© Vaibhav Yenare

P.T.O.

(continued)

#	Roman month	Meaning	Today
5	**Quintilis**	**fifth**	July
6	**Sextilis**	**sixth**	August
7	**September**	**seventh** (*septem*)	September
8	**October**	**eighth** (*octo*)	October
9	**November**	**ninth** (*novem*)	November
10	**December**	**tenth** (*decem*)	December

Pattern spotted!

September means "7th month" but is our 9th; December means "10th" but is our 12th. Compare Sanskrit *sapta* (7), *ashta* (8), *nava* (9), *dasha* (10) — the same Indo-European roots!

King Numa (about 713 BCE) adds January and February

Numa added **Ianuarius** (Janus, two-faced god of doors and beginnings) and **Februarius** (*februa*, purification rites), making a lunar-style year of **355 days**. Romans considered odd numbers lucky, so most months got 29 or 31 days; February — the month of purification and the dead — got the unlucky 28. Priests occasionally inserted an extra month (*Mercedonius*), often for political convenience, so by Caesar's time the calendar was about 3 months out of step with the seasons.

The Julian calendar (45 BCE)

Advised by the Alexandrian astronomer **Sosigenes**, **Julius Caesar** introduced a **solar** calendar:

$$365 \text{ days} + 1 \text{ leap day every 4 years} \Rightarrow \text{average year} = 365.25 \text{ days}.$$

To realign the seasons, 46 BCE was stretched to **445 days** — the "Year of Confusion". Caesar added 10 days to Numa's months, giving the lengths we still use. **Quintilis** was renamed **Iulius** (July) in 44 BCE to honour Caesar, and **Sextilis** became **Augustus** (August) in 8 BCE.

Month	Days	Month	Days	Month	Days	Month	Days
January	31	April	30	July	31	October	31
February	28/29	May	31	August	31	November	30
March	31	June	30	September	30	December	31

Myth-buster

A popular story says Augustus "stole" a day from February so that his month would be as long as Caesar's. There is **no Roman evidence** for this; the story was invented by a 13th-century writer (Sacrobosco). August already had 31 days in Caesar's reform.

Memory tricks. *"Thirty days hath September, April, June and November…"* — or the **knuckle method**: make two fists and count the months over the knuckles (31 days) and the gaps between them (30, or 28/29 for February). July and August fall on two neighbouring knuckles.

3. The Gregorian correction (1582)

The problem: 11 minutes a year

$$365.25 - 365.2422 = 0.0078 \text{ days} \approx 11 \text{ min } 14 \text{ s per year} \quad \Longrightarrow \quad 1 \text{ day every } \frac{1}{0.0078} \approx 128 \text{ years.}$$

The Council of Nicaea (325 CE) had fixed the March equinox on **21 March**; by the 1500s the real equinox had drifted to about **11 March**.

The fix

In **1582**, Pope **Gregory XIII** issued a reform based on work by the astronomers Aloysius Lilius and Christopher Clavius:

1. **Delete 10 days:** Thursday 4 October 1582 was followed by Friday 15 October 1582 — the weekday sequence was **not** broken.
2. **A new leap-year rule:**

> A year is a **leap year** if it is divisible by **4**,
> **except** century years (divisible by **100**), which are leap years **only if divisible by 400**.

Year	÷4?	÷100?	÷400?	Leap?
2024	✔	×	—	**Yes**
2026	×	—	—	No
1900	✔	✔	×	**No**
2000	✔	✔	✔	**Yes**
2100	✔	✔	×	**No**

There are 97 leap years in every 400 years, so

$$\text{average Gregorian year} = 365 + \frac{97}{400} = 365.2425 \text{ days,}$$

only about 26 seconds longer than the true year — an error of 1 day in about 3,300 years.

Who switched when?

Country	Switch	Days dropped
Italy, Spain, Portugal, Poland	1582	10
Britain and its colonies, including British India	**1752**	**11** (Wed 2 Sept → Thu 14 Sept)
Japan	1873	(switched from a lunisolar calendar)
Russia	1918	13 (the "October Revolution" of 1917 happened on 7 November, Gregorian!)
Greece	1923	13

British India used the Gregorian calendar for official purposes from 1752, alongside many regional Hindu, Islamic and Parsi calendars — as it still does. Since 1957 the Indian National Calendar has been official alongside the Gregorian.

4. Calendar calculations you can do in your head

A. Your birthday's weekday next year. $365 = 52 \times 7 + \mathbf{1}$, so a date moves forward **1 weekday** each year — or **2** if a 29 February falls in between.

Worked example

15 August 2026 is a Saturday. 15 Aug 2027: no 29 Feb in between ⇒ +1 ⇒ **Sunday**. 15 Aug 2028: 29 Feb 2028 in between ⇒ +2 ⇒ **Tuesday**.

B. Counting leap years. From 2001 to 2100: multiples of 4 from 2004 to 2100 number $\frac{2100-2004}{4}+1 = 25$; remove 2100 ⇒ **24**.

C. Julian Day Number (optional). Astronomers count days continuously from 1 January 4713 BCE: JD 2,451,545.0 = 1 January 2000, 12:00 UT. The number of days between two dates is just the difference of their JDs.

5. Clock time: whose noon is it?

Solar time and mean time

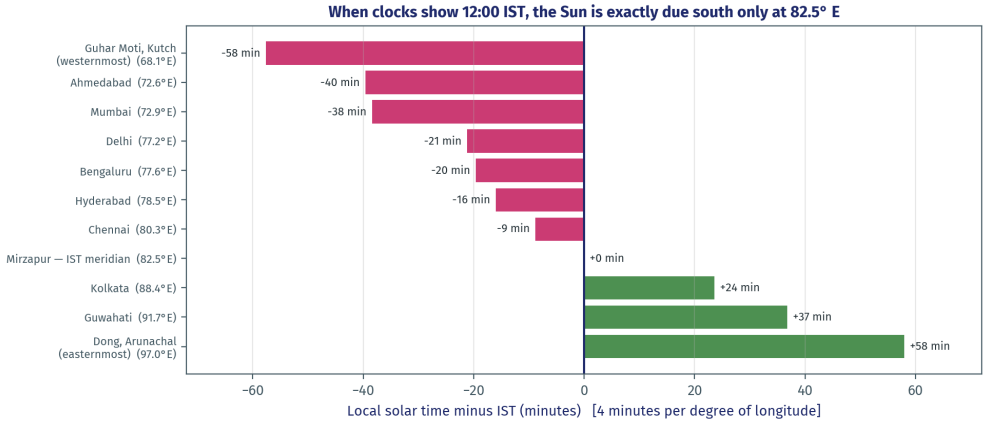
- A **sundial** shows *apparent solar time*: noon is when the Sun is due south (on the meridian).
- Because Earth's orbit is elliptical and its axis tilted, sundial time runs up to **16 min fast or 14 min slow** compared with a steady clock — the **equation of time**. Photograph the Sun at the same clock time all year and it traces a figure-8, the *analemma*.
- Clocks use **mean solar time** — an averaged, perfectly regular Sun.

Time zones and Indian Standard Time

Earth turns 15° per hour, so places 1° of longitude apart have local times **4 minutes** apart. Countries therefore use standard time zones.

- IST is based on the **82.5° E** meridian, which passes near **Mirzapur** (Uttar Pradesh).
- 82.5°/15° = 5.5 h, so **IST = UTC + 5 h 30 m**.
- Before IST (official from 1 September 1947; "Bombay Time" lingered until 1955) cities kept their own times, such as Bombay Time and Calcutta Time.
- The **CSIR–National Physical Laboratory, New Delhi** keeps IST with atomic clocks.

India spans about 29° of longitude (68° E to 97° E), so local solar time differs by about 2 hours from west to east — yet the whole country uses one clock.



How far each city's local solar time is from IST.

$$\text{Local mean time} = \text{IST} + 4\ \text{min} \times (\lambda - 82.5^\circ) \qquad (\lambda = \text{longitude, east positive})$$

Worked examples

1. Mumbai ($\lambda = 72.9^\circ$ E): $4 \times (72.9 - 82.5) = -38$ min. When IST is 12:00, Mumbai's local time is 11:22, so the Sun crosses Mumbai's meridian at about **12:38 PM IST** ($\pm$ the equation of time).

2. Kolkata ($\lambda = 88.4^\circ$ E): $4 \times (88.4 - 82.5) = +24$ min $\Rightarrow$ local noon $\approx$ **11:36 AM IST**. Assam's tea gardens historically kept their own "Chaibagan time", one hour ahead of IST.

3. Sunrise across India: Arunachal Pradesh (97° E) sees sunrise about $4 \times (97 - 68) \approx 116$ minutes earlier than Kutch (68° E) at similar latitudes.

The second itself

Since 1967 the **second** is defined by atomic physics: 9,192,631,770 oscillations of radiation from a caesium-133 atom. Earth's spin is slightly irregular, so **leap seconds** were occasionally added to keep clock time (UTC) in step with the Earth; they are due to be phased out by 2035.

Key points to remember

- Day, month (29.53 d) and year (365.2422 d) do not divide evenly — every calendar is a compromise.
- 12 months because a year $\approx$ 12.37 lunations; 30/31-day months come from Numa's 355-day year plus Caesar's 10 extra days.
- September–December mean 7th–10th: the Roman year once started in March.
- Julian: leap year every 4 years (365.25 d). Gregorian (1582): centuries are not leap years unless divisible by 400 (365.2425 d).
- Britain and India switched in **1752** (11 days dropped).
- IST = UTC + 5:30 (82.5° E); local time changes by 4 min per degree of longitude.

Check yourself

1. Is 2400 a leap year? Is 2200?
2. Which Roman month name means "eighth"? Which month is it now?
3. Guwahati is at 91.7° E. Is its local noon before or after 12:00 IST, and by how much?
4. Why was the Julian calendar wrong by about 1 day every 128 years?

Answers: 1. 2400 yes; 2200 no. 2. October — now the 10th month. 3. Before, by $4 \times (91.7 - 82.5) \approx 37$ min (about 11:23 AM IST). 4. Its year (365.25 d) is ~11 min longer than the tropical year, and 11 min $\times$ 128 $\approx$ 1 day.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §4.3 "Keeping Time" and §4.4 "The Calendar" — https://openstax.org/books/astronomy-2e/pages/4-3-keeping-time
- CSIR–National Physical Laboratory, Indian Standard Time — https://www.nplindia.org
- E. G. Richards, *Mapping Time: The Calendar and its History* (Oxford University Press).